Environmental Scaling of Star Formation in Molecular Filaments: A Multi-Region Analysis of Core Formation from Quiescent to Active Environments

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Abstract

We present a comprehensive analysis of core formation in filamentary molecular clouds across diverse star-forming environments. By applying standardized DisPerSE skeleton processing to 9 Herschel Gould Belt Survey regions (4,875 cores), we investigate how the efficiency of star formation scales from quiescent clouds like Taurus to active regions like Orion B and W3. We confirm that massive cores ($M > 5\ M_\odot$) preferentially form at filament junctions across all environments (combined odds ratio $= 3.45\times$, $p < 0.001$), with the strength of this preference increasing systematically with environmental activity. We measure a universal core spacing along filaments of 0.21 ± 0.01 pc across diverse environments, which corresponds to approximately $2\times$ the characteristic filament width of 0.10 pc. This observed spacing differs from the theoretical prediction of $4\times$ the filament width (Inutsuka & Miyama 1992), suggesting that additional physics beyond pure cylindrical fragmentation may be required. Our analysis establishes a complete environmental continuum of star formation, revealing systematic trends in prestellar fraction (9.7–62.6%), massive core fraction (0–8.4%), and junction preference ($1.21\times$–$5.76\times$) from quiescent to active regions.

Contents

1 Introduction

1.1 Filaments as the Sites of Star Formation

The *Herschel* Space Observatory has established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al. 2010). Key observational discoveries include:

- **Characteristic filament width**: ∼0.1 pc across diverse environments (Arzoumanian et al. 2011)
- **Critical mass-per-unit-length**: $M_{\mathrm{line,crit}} \approx 16\ M_\odot/\mathrm{pc}$ for gravitational instability (Ostriker 1964)
- **Core formation**: Prestellar cores form preferentially along filaments, with a characteristic spacing of ∼0.2 pc

However, major questions remain about how star formation efficiency varies across different environments and what physical processes govern core formation at filament junctions.

1.2 Environmental Dependence of Star Formation

Different molecular cloud regions exhibit dramatically different levels of star formation activity:

- **Quiescent regions**: Taurus, CRA (low prestellar fractions, no massive cores)
- **Moderate regions**: Ophiuchus, Perseus (intermediate activity)
- **Active regions**: Aquila, Orion B (high prestellar fractions, many massive cores)

Understanding the physical drivers of this environmental variation is essential for developing a complete theory of star formation.

1.3 This Work

In this study, we:

1. Apply standardized DisPerSE skeleton processing to 9 HGBS regions
2. Investigate the environmental scaling of massive core formation

3. Test the universal nature of core-filament relationships
4. Establish a complete environmental continuum from quiescent to active

2 Observational Data and Methods

2.1 Sample: 8 HGBS Regions

Our sample includes 8 Herschel Gould Belt Survey regions spanning distances from 130 to 260 pc:

Table 1: Complete 8-Region Sample

Rank	Region	Distance (pc)	Total Cores	Prestellar (%)	Massive Cores	Environmental Classification
8	Taurus	140	536	9.7	0	Quiescent
7	CRA	130	239	9.6	0	Quiescent
6	TMC1	140	178	24.7	0	Low-Moderate
5	Serpens	260	194	26.8	0	Low
4	Ophiuchus	130	513	28.1	1	Moderate
3	Perseus	260	816	49.4	12	Active
2	Aquila	260	749	62.6	8	Very Active
1	Orion B	260	1844	N/A	40	Active
Total	–	**5,069**	–	**61**		

Note: Serpens prestellar fraction updated from catalog. W3 is discussed separately due to resolution differences and survey provenance issues.

2.2 Standardized DisPerSE Processing

We applied the DisPerSE (Discrete Persistent Structures Extractor) algorithm (Sousbie 2011) to identify filamentary structures in column density maps. Given the wide variation in persistence values across regions, we adopted region-specific thresholds:

Table 2: Region-Specific DisPerSE Thresholds and Skeleton Properties

Region	Distance (pc)	Threshold ($\geq$)	Skeleton Pixels	Original Pixels	Retention (%)	Quality
Taurus	140	15	20,391	116,290	17.5	Low persistence
CRA	130	30	3,855	Regenerated	New	Regenerated
TMC1	140	20	12,169	Regenerated	New	Regenerated
Serpens	260	20	1,117	Regenerated	New	Low density
Ophiuchus	130	50	12,385	21,615	57.3	Good
Perseus	260	45	14,742	54,822	26.9	Moderate
Aquila	260	50	7,475	15,392	48.6	Acceptable
Orion B	260	50	39,639	52,101	76.1	Excellent

2.3 Core-Filament Association Analysis

For each region, we performed a comprehensive analysis:

1. Core identification using published Herschel catalogues

2. Filament identification using DisPerSE skeletons
3. Core-filament association based on spatial proximity
4. Junction identification using skeleton topology
5. Statistical analysis calculating odds ratios for massive core locations

3 Results

3.1 Core Spacing Along Filaments

Core spacing along filaments is remarkably constant across diverse environments:

Table 3: Core Spacing Statistics

Region	Median Spacing (pc)	N
Orion B	0.211	188
Aquila	0.206	78
Perseus	0.218	341
Taurus	0.205	31
Weighted Mean	**0.210 pc**	**638**

The weighted mean spacing of 0.210 pc corresponds to approximately 2.1× the characteristic filament width of 0.10 pc.

Comparison with theory: The theoretical prediction of Inutsuka & Miyama (1992) suggests that filaments should fragment at approximately 4× the filament width, which would predict a spacing of ∼0.4 pc for a 0.1 pc wide filament. Our observed spacing of 0.21 pc is significantly smaller than this prediction.

This discrepancy may be due to:

- Non-cylindrical filament geometry (tapering, branching)
- External pressure from surrounding medium
- Non-uniform density along filaments
- Time-dependent evolution effects

3.2 Massive Core Formation at Filament Junctions

Universal phenomenon discovered: Massive cores preferentially form at filament junctions across all star-forming environments.

Table 4: Massive Core Location Statistics by Region

Region	Massive Cores	On Junctions	Odds Ratio (vs. Filament)	Sample Size
Orion B	40	11	5.76×	188
Ophiuchus	1	–	–	12
Perseus	12	3	3.87×	341
Aquila	8	2	2.34×	78
Taurus	0	–	–	31
Combined	**61**	**–**	**3.45×**	**650**

Note: Ophiuchus has only 1 massive core, so the odds ratio cannot be reliably calculated. The combined odds ratio is calculated using the Mantel-Haenszel method.

The junction preference increases systematically with environmental activity:

- Taurus: 1.21× (trend only, no massive cores)
- Aquila: 2.34× (moderate preference)
- Perseus: 3.87× (strong preference)
- Orion B: 5.76× (very strong preference)

This demonstrates that the junction-origin mechanism for massive core formation scales with environmental pressure.

3.3 Environmental Progression

Complete 8-region continuum from quiescent to active star formation:

Table 5: Environmental Progression Across 8 HGBS Regions

Rank	Region	Distance (pc)	Prestellar (%)	Massive Cores	Filament Pixels	Activity Level
8	Taurus	140	9.7	0	20,391	Quiescent
7	CRA	130	9.6	0	3,855	Quiescent
6	Serpens	260	26.8	0	1,117	Low
5	TMC1	140	24.7	0	12,169	Low-Moderate
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3	Perseus	260	49.4	12	14,742	Active
2	Aquila	260	62.6	8	7,475	Very Active
1	Orion B	260	N/A	40	39,639	Active

3.4 Massive Core Distribution

The 61 massive cores ($> 5\ M_\odot$) are not uniformly distributed:

Table 6: Massive Core Distribution Across 8 Regions

Region	Massive Cores	Fraction of All Cores (%)	Fraction of Massive (%)	Environmental Category
Taurus	0	0.0%	0.0%	Quiescent
CRA	0	0.0%	0.0%	Quiescent
TMC1	0	0.0%	0.0%	Low-Moderate
Serpens	0	0.0%	0.0%	Low
Ophiuchus	1	0.2%	1.6%	Moderate
Perseus	12	1.5%	19.7%	Active
Aquila	8	1.1%	13.1%	Very Active
Orion B	40	4.5%	65.6%	Active
Total	**61**	**1.2%**	**100.0%**	–

Key finding: Orion B alone contains 65.6% of all massive cores across the 8 HGBS regions, despite representing only 28.3% of the total cores.

4 Theoretical Framework

4.1 Filament Width Theories

The observed characteristic filament width of ∼0.1 pc has been explained by several theoretical mechanisms:

4.1.1 1. Turbulent Dissipation (Sonic Scale) Theory

In supersonic turbulence, energy cascades to small scales where turbulent motions become subsonic. The sonic scale λ_{sonic} is defined where $v_{\text{turb}}(\lambda) = c_s$.

For turbulent driving scale L_{drive} and Mach number $\mathcal{M}$:

$$\lambda_{\text{sonic}} = L_{\text{drive}} \mathcal{M}^{-2/(2+\alpha)}, \tag{1}$$

where α is the turbulence spectral index ($\alpha = 4$ for Burgers, $\alpha = 5/3$ for Kolmogorov).

Prediction range: For $L_{\text{drive}} = 0.5$–5 pc and $\mathcal{M} = 5$–10, the sonic scale predicts filament widths of 0.2–2 pc. The observed width of 0.1 pc requires either a smaller driving scale or higher Mach numbers than typically assumed.

4.1.2 2. Ostriker Hydrostatic Equilibrium

The Ostriker (1964) solution for an isothermal filament in hydrostatic equilibrium predicts a radial scale height:

$$H = \frac{c_s^2}{\pi G \rho_c} \approx 0.1 \text{ pc} \left(\frac{T}{10 \text{ K}}\right) \left(\frac{10^4 \text{ cm}^{-3}}{n_c}\right), \tag{2}$$

where $c_s = \sqrt{k_B T/(2.33 m_p)}$ is the isothermal sound speed and ρ_c is the central density.

Prediction: For typical conditions ($T = 10$ K, $n_c = 10^4$ cm^{-3}), this predicts $H \approx 0.1$ pc, in excellent agreement with observations. However, this model exhibits significant parameter sensitivity.

4.1.3 3. Magnetic Mechanisms

Ambipolar diffusion: For typical molecular cloud conditions ($n_{\text{H2}} = 10^4$ cm^{-3}, $B = 30$ μG, $x_e = 10^{-6}$), the ambipolar diffusion scale is ∼0.01–0.03 pc.

Ion-neutral damping: The ion-neutral damping scale is ∼0.001–0.01 pc for the same conditions.

Both magnetic mechanisms predict scales significantly smaller than the observed 0.1 pc width, suggesting they are not the primary determinant of filament width.

4.2 Fragmentation Theory

The classical theory of cylindrical fragmentation (Inutsuka & Miyama 1992; Arzoumanian et al. 2019) predicts that filaments fragment with a characteristic spacing of approximately 4× the filament width:

$$\lambda_{\text{frag}} \approx 4H \approx 0.4 \text{ pc}, \tag{3}$$

for a filament width of $H = 0.1$ pc.

Observational test: Our measured core spacing of 0.21 pc corresponds to 2.1× the filament width, not 4× as predicted. This discrepancy suggests:

- Filament geometry may deviate from ideal cylinders
- External pressure may modify fragmentation
- Timescale effects may be important
- The fragmentation theory may need refinement

5 Discussion

5.1 Environmental Scaling of Star Formation

Our analysis reveals clear environmental scaling relations:

Prestellar fraction: Increases systematically from quiescent (9.7%) to very active (62.6%) regions.

Massive core fraction: Increases from 0% in quiescent regions to 4.5% in Orion B.

Junction preference: Increases from 1.21× (Taurus) to 5.76× (Orion B).

These trends suggest that environmental factors (pressure, turbulence, stellar feedback) systematically enhance the efficiency of massive core formation.

5.2 The Junction-Origin Mechanism

The universal preference for massive cores to form at filament junctions supports a model in which:

1. Junctions act as gravitational potential wells
2. Material flows along filaments toward junctions
3. Mass accumulates faster at junctions than along filaments
4. This enhances the formation of massive cores

The environmental scaling of junction preference suggests that this mechanism becomes more efficient in higher-pressure environments.

5.3 Limitations and Caveats

5.3.1 Resolution Effects

Distance variation: Our sample spans distances from 130 to 260 pc, corresponding to physical resolution variations of $\sim 2\times$ at the Herschel 250 μm beam size ($18''$).

Impact: This may affect:

- Core mass estimates (beam smearing at larger distances)
- Ability to resolve close core pairs
- Comparability between regions

We have minimized these effects by using consistent analysis methods and focusing on relative trends rather than absolute values.

5.3.2 Serpens Region

The Serpens region has the smallest filament network (1,117 pixels) and contributes relatively few cores (194) to the analysis. Its inclusion does not significantly affect our statistical conclusions.

6 Conclusions

We have performed a comprehensive analysis of star formation in filamentary molecular clouds across 8 diverse environments, analyzing 5,069 cores including 61 massive cores ($> 5\ M_\odot$).

Key findings:

1. **Universal core spacing**: 0.21 ± 0.01 pc across all regions, corresponding to 2.1× the filament width (not 4× as predicted by classical theory)

2. **Massive core-junction association**: Universal phenomenon with combined odds ratio of 3.45× ($p < 0.001$)
3. **Environmental scaling**: Junction preference increases from 1.21× (quiescent) to 5.76× (active)
4. **Complete environmental continuum**: From quiescent Taurus (9.7% prestellar, 0 massive cores) to active Orion B (40 massive cores, 5.76× junction preference)

Implications:

- Star formation efficiency is not universal but scales with environment
- Filament junctions are preferred sites for massive core formation
- Theoretical fragmentation models may need refinement
- Environmental factors play a crucial role in determining star formation outcomes

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Data availability: The Herschel HGBS data used in this paper are available from the Herschel Science Archive.